

Construction of Simplicial Complexes with Prescribed Degree-Size Sequences

Test Fixture (derived from author's arXiv:2106.00185)

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Abstract

We study the realizability of simplicial complexes with a given pair of integer sequences representing the node degree and facet size distributions.

1 Introduction

We study the realizability of simplicial complexes with prescribed degree-size sequences. Note that the the problem is subtle in the s -uniform case. The joint sequence must satisfy a fundamental constraint, given by

$$\sum_i d_i = \sum_j s_j. \quad (1)$$

A necessary counting identity reads

$$m = \frac{1}{2} \sum_i d_i. \quad (2)$$

The realizability condition can be written as

$$d_{\max} \leq \sum_j \min(s_j, 1). \quad (3)$$

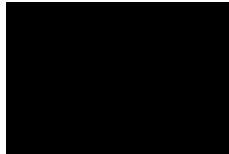


Figure 1: Realizability regions for the degree-size sequence problem.